Grounding LLMs at Scale

Enterprise RAG & Agentic Memory

With

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Amit Parmar
Data Solution Architect
Bank of America

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Some Interesting Concepts

Model Collapsing:

machine learning models degrade over time due to errors from training on synthetic data.

KV Cache:

KV cache, short for Key-Value cache, is a mechanism used in large language models (LLMs) to store the key (K) and value (V) tensors generated during the attention computations of transformer layers.

Model Requirement:

LLM RAM Calculator 2026 | Memory Requirements for Llama, Gemma, Qwen & More

01

CHAPTER 01

The RAG Revolution

Why RAG? The LLM Reality Check

LLMs are **powerful but fundamentally limited** — they hallucinate, have knowledge context limitation, and lack access to proprietary data.

RAG bridges this gap by grounding LLM responses in factual, retrieved context.

RAG is not optional — it's **a better choice** for enterprise AI.

69–88%

LLM hallucination rate on legal queries

Stanford RegLab & HAI Institute

1,200+ court cases involving AI hallucinations by 2026

The Problem in Numbers

10 → 73

AI hallucination court cases: 2023 → first 5 months of 2025

$145K

Q1 2026 sanctions for AI hallucinations — highest quarterly total

75%+

Hallucination rate on court core ruling questions

Source: AI Hallucination Rates & Benchmarks 2026, Stanford RegLab

FUN FACT: Even purpose-built legal AI tools fail — Lexis+ AI produced incorrect info 17%+ of the time, Westlaw AI-Assisted Research hallucinated 34%+

The RAG Evolution Spectrum

How RAG architectures have matured from simple retrieval to autonomous reasoning

NAIVE RAG

70–80%

Retrieval Precision

- Basic vector search
- Top-K retrieval
- No reranking
- No error correction

Best for: Prototypes, simple Q&A

MODULAR RAG

85–95%

Retrieval Precision

- Hybrid search (vector + BM25)
- Query rewriting
- Reranking
- Chunk optimization

Best for: Most production deployments

GRAPHRAG

90–97%

Retrieval Precision

- Knowledge graphs
- Cross-document reasoning
- Entity relationships
- Multi-hop queries

Best for: Complex reasoning tasks

AGENTIC RAG

92–99%

Retrieval Precision

- Adaptive retrieval
- Multi-step reasoning
- Tool use (APIs, SQL)
- Self-correction

Best for: Complex multi-step workflows

Complexity →

Source: Enterprise RAG Guide 2026

02

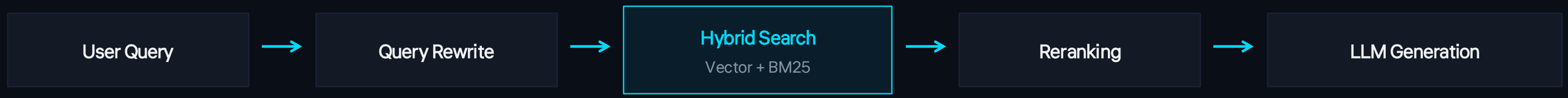
CHAPTER 02

Inside Enterprise RAG

Architecture patterns, challenges, and best practices

Enterprise RAG Architecture Patterns

Modular RAG Pipeline



Key Components

Hybrid Search

Combines dense (semantic) + sparse (BM25) retrieval for optimal precision and recall

Reranking

Secondary model scores retrieved documents, improving Top-K precision by 15–30%

Query Rewriting

Transforms ambiguous queries into optimized retrieval queries for better recall

Key Insight: Hybrid search combining dense (semantic) and sparse (BM25) retrieval **balances precision and recall** better than either approach alone. This is Elasticsearch's native superpower.

Enterprise Additions

- **Authentication** & Authorization
- **Metadata** filtering
- **Response** validation
- **Audit** logging
- **Central API** layer
- **Query** preprocessing

Sources: Enterprise RAG Guide 2026, RAG Architecture Guide

The Enterprise RAG Checklist

1 Data Preparation Strategy

- Intelligent chunking with overlap
- Metadata extraction & preservation
- Version control for documents
- Automated quality filtering

2 Vector Database Selection

- Scalability for growth
- Query performance (p99 latency)
- Security features & access controls
- Integration with existing ecosystem

3 Hybrid Retrieval Strategy

- Keyword + semantic search
- Reranking for precision
- Query reformulation
- User feedback loops

4 Security & Compliance

- RBAC / ABAC at document level
- Encryption at rest & in transit
- PII detection & redaction
- Audit logging across pipeline

5 Prompt Engineering

- Standardized prompt templates
- Source citation instructions
- Expected format specification
- Systematic testing & iteration

6 Governance

- Usage monitoring dashboards
- Source tracking & audit logs
- User satisfaction tracking
- Cost management & alerts

WARNING: 70% of RAG systems still lack evaluation frameworks. Don't be in the majority that ships blind. Measure hallucination rate, Precision@K, and end-to-end latency from day one.

Source: RAG Best Practices for Enterprise AI Teams

03

CHAPTER 03

AI Infrastructure

Elasticsearch: Beyond Text Search

From search engine to complete AI platform — the evolution that changes everything

Inference API

Unified API for connecting ML models — ELSER, OpenAI, Cohere, or custom models via Eland

Retrievers Framework

Composable search pipelines — standard Query DSL, kNN, RRF fusion, and semantic reranking

Vector Search

Native dense + sparse vector support with HNSW indexing, quantization, and similarity scoring

Document-Level Security

Role-based and attribute-based access control enforced at the database layer, not application

semantic_text Field Type

Auto-chunking, auto-embeddings, auto-dimension detection — just two lines of mapping code

Elastic Inference Service (EIS)

GPU-powered inference for ELSER and embedding models — always-on, auto-scaling, zero ops

One platform for search, vectors, and AI — no data movement, no sync complexity, no separate vector database to manage

Source: AI-Powered Search with Elasticsearch

Embedding Strategies with Elastic

Three paths, each with distinct operational implications — choose wisely

ELSER (Sparse)

- Runs natively on ES ML nodes
- Zero external dependencies
- Auto chunking enabled
- ~26 docs/sec throughput
- Memory-efficient sparse vectors

Limitations: English only, 512 token limit per field

Best for: English content, zero model management overhead

Third-Party APIs

- OpenAI, Cohere, Google
- Multilingual support
- Latest foundation models
- Per-token API pricing
- No infrastructure to manage

Limitations: External dependency, costly at scale

Best for: Multilingual, low-volume, no infra teams

Self-Hosted Dense

- Via Eland or custom deploy
- Full control & data sovereignty
- Domain-specific fine-tuning
- Lower per-query cost at scale
- No external dependencies

Limitations: GPU provisioning, model versioning ops

Best for: Data sovereignty, domain-specific vocab

Pragmatic approach: Start with ELSER or semantic_text with defaults → Measure relevance with a judgment list → Switch to dense only if results fall short on non-English or domain-specific queries

Source: Semantic Search with GPU Inference in Elastic

04

CHAPTER 04

Agentic Memory Architecture

Memory isn't just storage — it's the agent's architecture

The Four Pillars of Agentic Memory

Human cognition meets AI — four memory types inspired by how our brains actually work

WORKING MEMORY

In-Context

The active context window — the "RAM" of the agent. Everything currently in the prompt: system prompt, messages, tool outputs.

Analogy: Your computer's RAM

Lives in: LLM context window

EVENT MEMORY

Episodic

Specific past events and interactions — the "diary". What happened, when, with whom, and what was the outcome.

Analogy: Your personal diary

Lives in: Vector DB

KNOWLEDGE BASE

Semantic

Facts, definitions, world knowledge — the "encyclopedia". General knowledge the agent uses to reason about new problems.

Analogy: Wikipedia in your head

Lives in: Vector DB

SKILL MEMORY

Procedural

Skills, rules, workflows — the "muscle memory". How to perform tasks that become automatic through repetition.

Analogy: Riding a bicycle

Lives in: System prompts, agent code

Together these transform agents from stateless chatbots into systems that **learn, adapt, and improve over time**

Note: mem0 is one of the good solution in today's time to try and explore the agentic memory layer!

Sources: Memory for Agents (LangChain), Types of AI Agent Memory (Atlan)

Memory + RAG: The Perfect Partnership

RAG is the **bridge between short-term and long-term memory**. It's not about adding more context — it's about adding the **right** context.

2 relevant paragraphs

beat 20 loosely related pages — every time

RAG Best Practices for Memory

Targeted Retrieval

Only the most relevant chunks — not bulk loading entire documents

Just-in-Time Recall

Fetch context only when needed, not carrying everything forward

Relevance-Aware Strategies

Top-k semantic, RRF fusion, tool loadout filtering

The Context Window Myth

"Larger context windows don't remove the need for RAG."

Even 1M-token context windows suffer from:

- Context pollution (noise drowns signal)
- Higher latency and cost
- Lost in the middle problem

RAG = precision targeting. Context stuffing = carpet bombing.

"RAG isn't about adding more context; it's about adding the right context."

Source: Agentic Memory Management with Elasticsearch

The "Severance" Demo — Selective Memory in Action

Watch how the same agent responds differently based on who's asking

AGENT: MARK

Peter ("Outtie")

Role: outtie_access

"What are my work tasks for today?"

→ ES filters to "outtie" memories only
→ Returns: Work tasks, meetings
→ Innie data: INVISIBLE

Janice ("Innie")

Role: innie_access

"What are my work tasks for today?"

→ ES filters to "innie" memories only
→ Returns: Inner work, projects
→ Outtie data: INVISIBLE

The Magic: Zero Application Code

Elasticsearch's Document-Level Security handles everything. The agent code just passes the user's credentials through. No if-statements, no filtering logic, no bugs.

FOR THE FANS: "The work is mysterious and important... and now it's also securely isolated per tenant." — Mark (probably)

Source: Agentic AI Memory Management with Elasticsearch

05

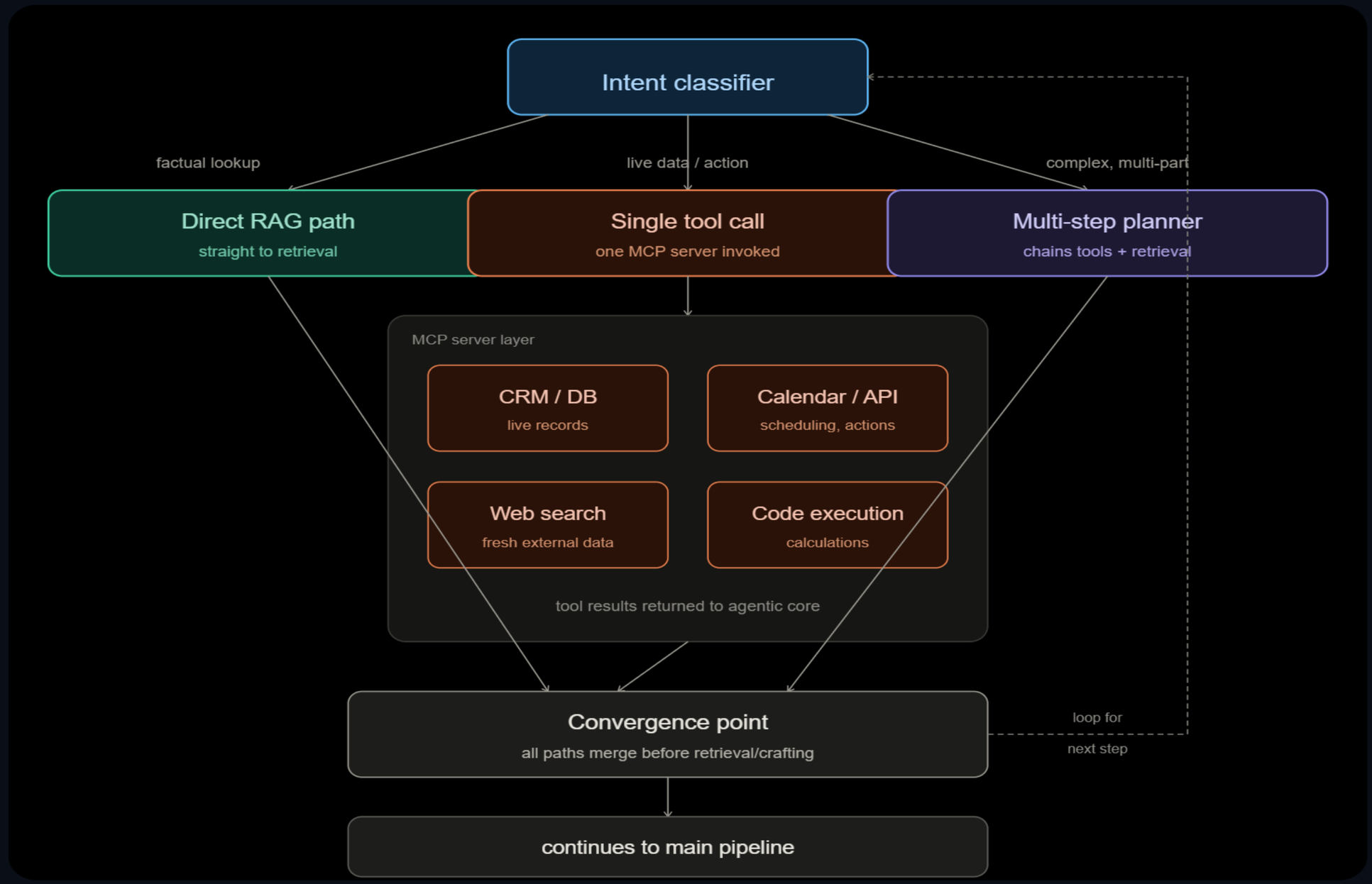
CHAPTER 05

Building for Production

Real-world implementation roadmap and key takeaways

Enterprise RAG Implementation Roadmap

Enterprise RAG Implementation Roadmap



Key Takeaways — Grounding LLMs at Scale

1 **RAG is essential, not optional** — LLMs without grounding are liabilities in enterprise environments

2 **Start with Modular RAG** — it's the sweet spot for most enterprises (85–95% precision)

3 **Elasticsearch = Search + Vectors + AI** — one platform eliminates data movement complexity

4 **Agentic memory transforms bots** — episodic + semantic + procedural = learning systems

5 **Evaluation is non-negotiable** — 70% of RAG systems lack it. Be in the 30% that succeed.

> *"Good context engineering looks like good systems design: smaller, sufficient contexts, controlled interfaces, and clear separation of concerns."*

Ready to Ground Your LLMs?

- Start with Elasticsearch's **semantic_text** field + ELSER for instant semantic search
- Add **Document-Level Security** for multi-tenant agentic memory
- Measure everything — **hallucination rate, Precision@K, latency** — from day one

Thanks !
Grounding LLMs at Scale

Enterprise RAG & Agentic Memory with Elasticsearch

Questions? Let's discuss!

Share your thoughts and let's Connect!